

## Random Variable

### Reading Objectives:

After going through this reading, you will be able to define random variables and learn about the types of random variables with examples. You will also be able to introduce the concept of expectation and its properties.

### Main Reading Section:

#### 1. Random Variable

When a random experiment is performed, we are often not interested in all of the details of the experimental result but only in the value of some numerical quantity determined by the result. For instance, in tossing dice, we are often interested in the sum of the two dice and are not really concerned about the values of the individual dice. That is, we may be interested in knowing that the sum is 7 and not be concerned over whether the actual outcome was (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), or (6, 1). These quantities of interest that are determined by the result of the experiment are known as **random variables**.

**Example 1.** Suppose that an individual purchases two electronic components, each of which may be either defective or acceptable. In addition, suppose that the four possible results—(d, d), (d, a), (a, d), and (a, a)—have respective probabilities 0.09, 0.21, 0.21, and 0.49 [where (d, d) means that both components are defective, (d, a) means that the first component is defective and the second acceptable, and so on]. If we let  $X$  denote the number of acceptable components obtained in the purchase, then  $X$  is a random variable taking on one of the values 0, 1, and 2 with respective probabilities

$$P\{X = 0\} = 0.09$$

$$P\{X = 1\} = 0.42$$

$$P\{X = 2\} = 0.49$$

If we were mainly concerned with whether there was at least one acceptable component, we could define the random variable  $I$  as follows:

$$I = \begin{cases} 1 & \text{if } X = 1 \text{ or } 2 \\ 2 & \text{if } X = 0 \end{cases}$$

#### 2. Types of Random Variables

Random variables whose set of possible values can be written either as a finite sequence  $x_1, x_2, \dots, x_n$  or as an infinite sequence  $x_1, x_2, \dots$ , are said to be **discrete**.

However, there also exist random variables that take on a continuum of possible values. These are known as **continuous** random variables.

**Definition 1:** For a discrete random variable  $X$ , we define the **probability mass function**  $p(a)$  of  $X$  by  $p(a) = P\{X = a\}$ . The probability mass function  $p(a)$  is positive for at most a countable number of values of  $a$ . That is, if  $X$  must assume one of the values  $x_1, x_2, \dots$ , then

$$p(x_i) > 0, \quad \text{for } i = 1, 2, \dots$$

$$p(x) = 0, \quad \text{for all other values of } x.$$

Since  $X$  must take on one of the values  $x_i$ , we have  $\sum_{i=1}^{\infty} p(x_i) = 1$

### Expectation

#### Definition 2:

If  $X$  is a discrete random variable taking on the possible values  $x_1, x_2, \dots$ , then the expectation or expected value of  $X$ , denoted by  $E[X]$ , is defined by

$$E[X] = \sum_i x_i P\{X = x_i\}$$

Suppose that  $X$  is a continuous random variable with probability density function  $f$ . Since, for a small value of  $dx$ ,  $f(x)dx \approx P\{x < X < x + dx\}$ . Hence, it is natural to define the expected value of  $X$  by

$$E[X] = \int_{-\infty}^{\infty} x f(x) dx$$

### Properties of the expected value

If  $a$  and  $b$  are constants, then  $E[aX + b] = aE[X] + b$

$$E[X^n] = \begin{cases} \sum_x x^n p(x), & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} x^n f(x) dx, & \text{if } X \text{ is continuous} \end{cases}$$